

The Research Essay: Analyzing the evolution of behavioral sentiment in relation to narrative comprehension

Every day we comprehend narratives—reading news articles, following the gossip our friends tell us, listening to our professor or boss, or watching dozens of episodes of our favorite binge-worthy television series. In many of these scenarios, we come to form opinions, likes or dislikes, around the people or topics we learn about in these narratives. For example, if your friend dishes to you that her uncle’s mailman puts glitter bombs in the mailboxes of residents he dislikes, depending on how your friend’s framing of the story and her level of disclosure, you may come to think the mailman is childish, an awesome prankster, deeply unprofessional, or just a guy trying to make his job more exciting. Depending on the amount of context you have or how well you followed the story, you may or may not come to a different opinion of the mailman. In fact, upon reciting the story, your friend may now have a new opinion of the mailman. This illustrates the question I hope to study: how do narratives shape our understanding of the social world, and what neural mechanisms support this process?

I am collaborating with post-doc Hayoung Song and PI Monica Rosenberg to address these questions using behavioral and brain imaging data collected while people watched a social narrative. In this functional magnetic resonance imaging (fMRI) experiment, participants watched an episode of *This is Us* with the scenes scrambled. Participants pressed the “aha!” button whenever they realized something new about the story. After every five scrambled scenes, participants expressed their opinions about each character. Audio data and fMRI neural data were collected for this task.

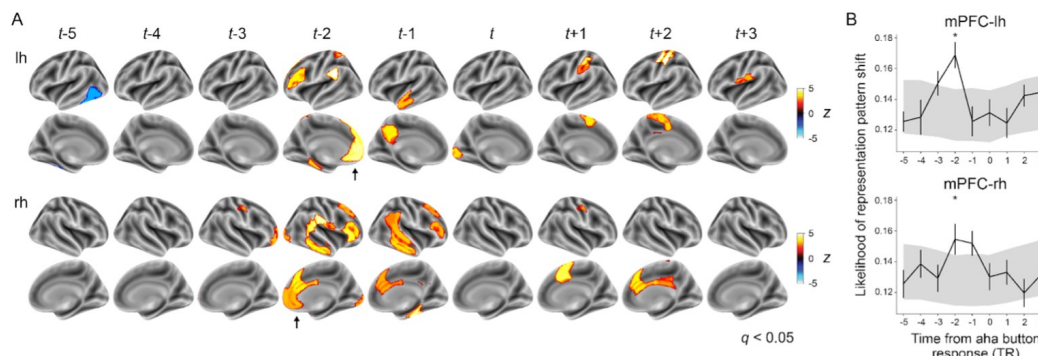


Figure 1. Likelihood of representation pattern shifts time-aligned to aha button responses¹

Ongoing analyses revealed a neural indication of moments of insight/“aha” and a common “big aha” moment where participants understood the connections between all characters and the plot. We found that comprehension of each character’s storyline

¹ Song, Hayoung. “Comprehension of causal structure in narratives.” 24 June. 2024. CAB Lab, University of Chicago, unpublished.

strengthens as the story unfolds. So, with this data, I ask: does understanding a character correlate with liking or disliking them? What impact does comprehension have on sentiment towards a character? These questions have implications for how our emotional attitudes neurally and behaviorally change through storytelling.

My analysis and addition to the existing research on the collected data will unfold in roughly six parts with mentorship from Drs. Hayoung Song and Monica Rosenberg. First, in compliance with the IRB, I will collect additional behavioral data to understand how people break narratives into events. A paper entitled, “Discovering event structure in continuous narrative perception and memory,”² coined research on how humans segment experiences into discrete events in their minds; those segmentations are called event boundaries. Some people have asked if these event boundaries are correlated, or equal to, moments of insight. By collecting this data, we can qualify existing findings about the “aha”/insight moments identified in the dataset: are “aha” moments or moments of insight simply just participant-identified event boundaries? The data are also vital for analyzing how narratives shape our social percepts because they allow us to divide the narrative into units for analysis. After collecting this data, I will determine whether or not what people *tell* us about a character aligns with what their free speech suggests they think, comparing quantified attitudes towards characters with sentiment analysis of participants’ speech. This analysis will gauge how positively or negatively participants feel about each character over time using machine learning (ML) and natural language processing techniques. I will employ tools like the Natural Language Toolkit (NLTK), and learn and explore others like SpaCy or BERT modeling. I will conduct sentiment analysis on This Is Us characters using the text I transcribed from the raw data (audio of participants’ expressed character opinions). I hypothesize a correlation between verbal sentiment progression and surveyed attitudes, though the strength of this correlation remains unclear.

Next, I will examine the neural basis for changes in attitude towards characters by identifying patterns of fMRI activity that shift with participant attitudes. Specifically, I will test (a) whether people with similar feelings towards a character share similar neural patterns upon seeing the character and (b) if people with similar trajectories in character opinion (e.g., liking them more vs. less over time) show synchronized neural activity in brain regions implicated in social perception and evaluation. I will measure attitude with both self-reported ratings and sentiment from free-speech data.

Additionally, I hope to conduct tonal sentiment analysis on the raw audio data by comparing neural data from watching character scenes with speech neural data to support or complicate my findings. Tone can change sentence meaning significantly, so, if results differ, I may re-perform prior sentiment analysis with audio data. This tonal analysis would, both, help to qualify my findings earlier in the process and be an opportunity to practice machine learning techniques and possibly build my own model for impactful use.

As a Computer Science and Cognitive Science student, my proficiency in Python and ability to read documentation have aided my start at the CAB Lab, allowing me to quickly pick up new languages, like Matlab. Furthermore, I have extensive knowledge of the dataset through my beginnings as a data labeler and audio transcriber for

2 Baldassano, Christopher et al. “Discovering Event Structure in Continuous Narrative Perception and Memory.” *Neuron* vol. 95,3 (2017): 709-721.e5. doi:10.1016/j.neuron.2017.06.041

Rhea Madhogarhia
University of Chicago

behavioral and fMRI experiments. With this existing understanding of the data, I'm excited for the opportunity to uncover new findings and explore the potential of the robust dataset already collected. After all, it was the time I spent transcribing that sparked my interest in shifts in tone and sentiment in participant responses.

My motivation to interrogate the dataset in this way comes from a deep-rooted interest in the brain and cognition in relation to psychotherapy. Psychotherapy relies on verbalization and narrative construction to explore thoughts and feelings. The neurological confirmation of psychotherapy's effectiveness is an emerging dialogue in clinical psychology; given the data, I feel personally and uniquely positioned to engage in this research discussion from a social cognitive neuroscience perspective.

Since joining the CAB Lab, I've discovered a love for research. Conversations with Dr. Song and Dr. Rosenberg reminded me of why I pursued cognitive science: to pose mind-boggling questions, have fun and stimulating conversations, and realize my ideas by applying both theoretical and practical computational skills to studies of the mind. For me, research has been the most fulfilling avenue to learn outside of the classroom: learning by doing. This project's interdisciplinary nature exposes me to skills relevant to both industry work and academia: the finetuning and modification of AI models is an important technical skill to have in the responsible/ethical tech realm, working with fMRI data is essential to neuropsychology research, and understanding the socio-cognitive, and neural connections within my research is vital to a more holistic and nuanced practice of clinical therapy. I hope that in all my academic and professional life, research plays an important role in my work. Whether working in the corporate engineering space, tech policy field, or in an artificial intelligence or cognitive neuroscience lab, I aspire for research to award me a deeper, more evidenced, and nuanced perspective.

Works Cited

Baldassano, Christopher et al. "Discovering Event Structure in Continuous Narrative Perception and Memory." *Neuron* vol. 95,3 (2017): 709-721.e5.
doi:10.1016/j.neuron.2017.06.041

Song, Hayoung. "Comprehension of causal structure in narratives." 24 June. 2024. CAB Lab, University of Chicago, unpublished.